Functional Requirements Document (FRD)

1. Project identification

Item	Information
Project title	KOHA-Lite Inventory Management for Little Free Libraries
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Course/section	01:198:437:01
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Version	2.1

2. Business problem and project purpose

Participating Little Free Libraries in New Jersey need a shared database for recording the circulation and location of their books. The libraries currently rely on an honor system, so directors cannot reliably determine which recorded books are available, which copies are on loan, where a returned copy is located, or which titles are in demand.

KOHA-Lite models circulation within participating libraries. Registered patrons and authorized directors record every checkout and return. Inventory represents recorded activity rather than independently verified physical inventory. Directors can reconcile differences between recorded and observed inventory for libraries assigned to them, and every reconciliation retains an attributable audit record.

The project will provide a normalized MySQL database that tracks physical book copies, circulation, donations, visits, director assignments, inventory corrections, and operational reports. It will not include a public website, mobile application, geographic map, cameras, automated physical surveillance, or production identity management.

3. Scope

In scope	Out of scope
Registered fictional users with patron, donor, or director roles	Anonymous or unrecorded borrowing, returns, or donations
Participating libraries and their operational status	Geographic map rendering, route planning, or GPS tracking
Book titles and individually identified physical copies	Book-condition grading, valuation, pricing, fines, or sales
Recorded availability and current library for every copy	Guaranteeing that recorded inventory exactly matches physical inventory
Checkout and mandatory-return workflows	Permanent taking of a checked-out copy
Returns to the originating or another participating library	Waiting lists, reservations, or interlibrary shipping
Donation submission and director acceptance	Purchasing and supplier management
Explicit patron library visits	Passive surveillance or automatic visit detection
Director reconciliation of recorded inventory drift	Public-facing web or mobile application development
Role-based access, audit history, reports, and database tests	Production authentication, notifications, and enterprise backup automation

4. Stakeholders, roles, and entitlements

One user may hold multiple roles. A director may manage one or more assigned libraries. Director authority is scoped to assigned libraries; it is not automatically global.

Role	Responsibilities	Database needs
Patron	Records personal checkouts, returns, and visits; reviews personal history	Available-book catalog, own active loans, own borrowing history, own visit history
Donor	Submits donations and reviews aggregate demand information	Donation submission/status and aggregate demand reports
Director	Maintains assigned libraries, records circulation for patrons, accepts donations, and reconciles inventory	Assigned-library inventory, controlled procedures, operational reports, and audit history

Permission matrix

Operation	Patron	Donor	Director
View available books	Yes	Yes	Yes
View aggregate demand	Yes	Yes	Yes
View personal borrowing and visit history	Own records	Only when also a patron	Own records; another patron only for an assigned-library operational task
Record checkout or return	Own activity	Only when also a patron	For patrons at assigned libraries
Record a visit	Own visit	Only when also a patron	For patrons at assigned libraries
Submit a donation	Yes	Yes	Yes
Accept or reject a donation	No	No	Assigned receiving libraries only
Register a physical copy	No	No	Assigned libraries only
Reconcile inventory drift	No	No	Assigned libraries only
Change an assigned library's operational status	No	No	Assigned libraries only
Modify or delete audit history	No	No	No

Roles above are business roles stored in the schema. MySQL roles and grants provide a second enforcement layer for representative database users; the project does not require one MySQL account per patron.

5. Functional requirements

ID	Functional requirement	Priority	Related data/entities
FR-01	An authorized director shall register an individual physical book copy at a library assigned to that director.	High	UserAccount, DirectorLibrary, BookTitle, BookCopy, Library
FR-02	The system shall list available copies at a selected active library, including title, author, category, and available-copy count.	High	Library, BookTitle, BookCategory, BookCopy
FR-03	A registered patron, or a director acting for that patron at an assigned library, shall check out an available copy. A successful checkout shall create one active loan and mark the copy on loan in one transaction.	High	UserAccount, UserRole, BookCopy, Loan, Library
FR-04	A registered patron, or a director acting for that patron, shall return a copy with an active loan. A successful return shall close the loan, mark the copy available, and set its current location to the selected participating return library in one transaction.	High	Loan, BookCopy, Library
FR-05	A registered user shall submit a donation to a selected active library, and an assigned director shall accept or reject it. Accepted physical copies shall be individually registered and linked to the donation.	Medium	Donation, DonationItem, BookTitle, BookCopy, Library
FR-06	A patron, or an assigned director acting for the patron, shall record an explicit library visit with its library, timestamp, and recording actor.	Medium	Visit, UserAccount, Library

ID	Functional requirement	Priority	Related data/entities
FR-07	A patron shall retrieve only that patron's borrowing and visit history. A director may retrieve patron activity only when it concerns a library assigned to that director.	High	Loan, Visit, DirectorLibrary
FR-08	The system shall report demand as the number of distinct patrons who checked out each title during a required date range, optionally filtered to one library, ordered from greatest to least demand.	Medium	Loan, BookCopy, BookTitle, Library
FR-09	An assigned director shall reconcile a recorded copy's location or status when observed inventory differs, supplying a nonblank reason and preserving the prior and new values.	High	InventoryAdjustment, BookCopy, DirectorLibrary
FR-10	The system shall reject unauthorized actions, inactive users, inactive libraries, unavailable-copy checkouts, returns without active loans, and invalid state transitions without leaving partial circulation or inventory changes.	High	All transactional entities
FR-11	The system shall create append-only audit evidence for successful state-changing business operations, including actor, event, affected record, timestamp, and relevant context.	High	AuditEvent
FR-12	An assigned director shall mark that library active or inactive through a controlled operation. Inactive libraries shall retain history but shall not accept new checkout, return, visit, donation, or copy-registration activity.	Medium	Library, DirectorLibrary, AuditEvent
FR-13	A repeated checkout or return request that conflicts with the current state shall be rejected without creating another loan, return, or inventory change.	High	Loan, BookCopy, AuditEvent

6. Data requirements

Data subject	Information to store	Identifier
User account	Synthetic display name, synthetic email, active status, creation time	UserID
Role assignment	User and business role	UserID + RoleCode
Library	Name, address, operational status	LibraryID
Director assignment	Director and managed library	UserID + LibraryID
Book category	Category name and description	CategoryID
Book title	ISBN when available, title, author, category	BookTitleID
Book copy	Title, current library, status, registration source and time	BookCopyID
Loan	Copy, patron, checkout library/actor/time, return library/actor/time, state	LoanID
Donation	Donor, receiving library, submission state, decision actor/time	DonationID
Donation item	Donation, title, offered quantity, accepted quantity	DonationItemID
Visit	Patron, library, visit time, recording actor and method	VisitID
Inventory adjustment	Copy, director, prior/new status and location, reason, time	AdjustmentID
Audit event	Actor, event type, entity, record identifier, timestamp, outcome, context	AuditEventID

All identities and contact values used for implementation and demonstration are synthetic. Price is not stored because the system does not perform valuation, purchasing, fines, insurance, or sales. Availability and quantity are derived from individual copy records rather than maintained as an independently editable quantity.

7. Business and validation rules

ID	Business rule	Planned database enforcement
BR-01	Every persistent record has a unique identifier; role and assignment pairs cannot be duplicated.	Primary keys and UNIQUE constraints
BR-02	Every physical copy belongs to exactly one book title and has at most one current library.	NOT NULL title FK; nullable library FK only for retired/lost states
BR-03	A copy may have at most one active loan.	Generated active-copy key with UNIQUE constraint plus controlled procedure
BR-04	Only an AVAILABLE copy at the selected active library may be checked out.	Transactional procedure, locking read, status checks
BR-05	A checkout and its copy-status change succeed or fail together.	InnoDB transaction with rollback handler
BR-06	A return must reference an active loan, and a returned copy becomes AVAILABLE at the participating return library.	Transactional procedure and state validation
BR-07	A patron must be active and have the PATRON role before a checkout, return, or visit is recorded for that patron.	Role-check function and controlled procedures
BR-08	A director may register copies, accept donations, record patron activity, or reconcile inventory only for an assigned library.	Director-assignment function and controlled procedures
BR-09	Donation offered quantity must be positive; accepted quantity must be between zero and offered quantity.	CHECK constraints and acceptance procedure
BR-10	Demand is COUNT(DISTINCT PatronUserID) over successful checkouts in a required date range.	Parameterized report query
BR-11	An inactive library retains existing records but cannot receive new operational activity.	Library-status checks in controlled procedures
BR-12	Inventory reconciliation requires an actor, affected copy, prior/new values, and nonblank reason.	NOT NULL/CHECK constraints and reconciliation procedure
BR-13	Audit events are append-only.	Privileges plus triggers that reject UPDATE and DELETE
BR-14	Completed circulation history is not overwritten or hard-deleted.	No direct DML grants; append/close workflow through procedures
BR-15	A failed or repeated circulation request leaves the database in its prior consistent state.	State checks, unique active-loan constraint, rollback handler

8. Reports and queries

ID	Report/query	Inputs	Required result
RQ-01	Available catalog	Library ID; optional category	Library, title, author, category, available-copy count; title order
RQ-02	Patron active loans	Authorized patron identity	Copy ID, title, checkout library and time; oldest checkout first
RQ-03	Patron borrowing history	Authorized patron identity; optional date range	Copy/title, checkout and return timestamps, originating and return libraries, loan state
RQ-04	Title demand	Required start/end timestamps; optional library	Title, distinct-patron count, checkout count; descending distinct-patron count then title
RQ-05	Patron visit history	Authorized patron identity; optional date range	Library, visit timestamp, recording method; newest first
RQ-06	Assigned-library inventory	Director identity and assigned library	Copy, title, recorded location, copy status, active-loan indicator
RQ-07	Donation status	Donor identity or assigned director	Donation, receiving library, items, offered/accepted counts, status and decision time
RQ-08	Audit history	Authorized director and assigned library or affected record	Actor, event type, timestamp, affected entity/ID, result and context

9. Assumptions and constraints

- All circulation, returns, donations, and explicit visits covered by the system are recorded; anonymous activity is outside the modeled workflow.
- A checked-out copy is expected to be returned. The project does not impose a due date, overdue fine, or notification because none was requested.
- A return may occur at a participating library different from the checkout library.
- Recorded inventory can drift from observed physical inventory; assigned directors correct drift through an audited reconciliation operation.
- One user may have multiple business roles, and one director may be assigned to multiple libraries.
- Library visits are explicit user/director-recorded events, not inferred from loans and not captured through surveillance.
- The database is implemented for MySQL 8.0 or later using InnoDB and UTF-8 (`utf8mb4`).
- The implementation uses fictional sample identities and does not contain real personal information or credentials.
- A public UI, production authentication service, production backup automation, and physical inventory verification are outside scope.

10. Acceptance criteria

Requirement	Observable acceptance criterion
FR-01	An assigned director registers a copy successfully; an unassigned director's attempt is rejected with no copy inserted.
FR-02	With a known fixture, the selected library catalog returns the expected titles and available-copy counts and excludes on-loan copies.
FR-03	Checkout of an available copy creates exactly one active loan and changes the copy to ON_LOAN in one transaction. Checkout of an unavailable copy changes nothing.
FR-04	Return closes the active loan and makes the copy available at the chosen participating library. A duplicate return changes nothing.
FR-05	A donor submits a donation; an assigned director accepts it; accepted counts and linked individual copies match the approved quantities.
FR-06	Patron self-check-in and director-recorded check-in create visits with the correct patron, library, actor, time, and recording method.
FR-07	A patron can retrieve that patron's history and cannot retrieve another patron's private history through the protected interface.
FR-08	For a fixture containing known checkouts, the demand report returns the expected distinct-patron count for the supplied date range and library filter.
FR-09	An assigned director changes recorded location/status with a reason; the copy and immutable adjustment history show the expected prior and new values.
FR-10	Unauthorized, invalid-state, and inactive-library operations return errors and leave related tables unchanged.
FR-11	Each successful state-changing operation creates an audit event containing actor, event type, timestamp, entity/ID, and context. UPDATE or DELETE of an audit event is rejected.
FR-12	After a library becomes inactive, history remains queryable while new operational transactions for that library are rejected.
FR-13	Two requests for one available copy result in at most one active loan; retrying a completed return creates no extra return or inventory increment.

Traceability note: Every FR requirement is mapped in the TDD to tables, constraints, procedures, reports, tests, and reproducible validation evidence. The full executable definition is maintained in `database/`, and the authoritative build order is `database/manifest.txt`.

11. Revision history and feedback disposition

Document revision history

Version	Revision	Summary
1.0	Initial draft	Established the Little Free Library topic, initial scope, three functional requirements, three business rules, three reports, assumptions, and acceptance criteria.
2.0	Requirements revision	Reworked the FRD after instructor feedback and project-scope clarification. Added explicit workflows, physical-copy tracking, permissions, reporting definitions, auditability, transaction rules, and testable acceptance criteria.
2.1	Feedback traceability revision	Added this revision history and documented which feedback was adopted, adapted, or not adopted, including the reason for each decision. No business behavior changed from version 2.0.

Feedback disposition

`Adopted` means the recommendation was added substantially as proposed. `Adapted` means the underlying concern was accepted but implemented in a form chosen for this project's scope. `Not adopted` means the recommendation or implied feature was deliberately excluded, with the reason recorded below.

Feedback topic	Disposition	Revision and rationale
Clarify who records borrowing and returns	Adopted	FR-03 and FR-04 allow a registered patron to record personal activity or an assigned director to record it for the patron. This makes every circulation event attributable.
Require patron registration	Adopted	Anonymous checkout is outside scope. Patron-role and active-user validation are explicit in BR-07 and the checkout/return workflows.
Decide whether activity may occur outside the system	Adopted	Anonymous or unrecorded borrowing, returns, and donations are excluded from the modeled workflow so recorded circulation remains auditable.
Decide whether a checked-out book must be returned	Adopted	A checkout creates an active loan that is expected to be returned. Permanent taking, due dates, fines, and notifications are outside scope.
Decide whether a copy may be returned elsewhere	Adopted	FR-04 permits return to another participating active library and updates the copy's current location. This supports the real movement of copies without adding shipping or transfer workflows.
Distinguish recorded from verified physical inventory	Adapted	The system reports recorded inventory, not independently verified physical inventory. FR-09 adds director reconciliation when observed inventory differs, with prior/new values and a reason.
Convert stakeholder descriptions into entitlements	Adopted	Section 4 now includes a permission matrix defining what patrons, donors, and directors may view or change. Director authority is limited to assigned libraries.
Clarify multiple roles and director assignments	Adopted	One user may hold multiple roles and one director may manage multiple libraries through explicit role and assignment relationships.
Decide whether roles are business roles or database roles	Adapted	Both layers are retained: schema-level business roles express requirements, while representative MySQL roles, grants, views, and procedures demonstrate technical enforcement. One MySQL account per patron is not required.
Rewrite and expand the functional requirements	Adapted	The three vague original requirements were replaced by thirteen numbered, testable requirements covering registration, catalog access, circulation, donations, visits, reports, reconciliation, authorization, audit, library status, and retries. IDs were reorganized to preserve a coherent final requirement set rather than copying the proposed list verbatim.
Separate book titles from physical copies	Adopted	`BookTitle` stores shared title/author/category information and `BookCopy` stores each physical copy, location, status, and registration source. This supports multiple copies and independent circulation.

Feedback topic	Disposition	Revision and rationale
Explain or remove price	Adopted	Price was removed because the project performs no purchasing, valuation, fines, insurance, or sales. Keeping it would add unused and unexplained data.
Decide whether quantity is stored or calculated	Adopted	Quantity and availability are calculated from individual `BookCopy` rows. No independently editable quantity is stored, preventing disagreement between a count and copy-level state.
Clarify library visits	Adopted	Visits are explicit `Visit` events rather than being inferred from circulation. A patron may self-record a visit, or an assigned director may record it for the patron. No passive tracking is introduced.
Define checkout, return, donation, and reconciliation workflows	Adopted	Requirements, business rules, and acceptance tests now state the actor, preconditions, state changes, rejection behavior, and audit outcome for each retained workflow.
Replace foreign-key enforcement for availability	Adopted	Availability is enforced through controlled transactional logic and copy state. Foreign keys are used only for referential integrity, not as a substitute for workflow validation.
Address simultaneous checkout requests	Adopted	BR-03 through BR-05 require at most one active loan, a locking read, a unique active-copy key, and a transaction that changes the loan and copy together.
Make reports precise	Adopted	Section 8 now defines inputs, output fields, aggregation, filters, and ordering for eight operational reports.
Define high demand and its time window	Adapted	High demand is defined as the number of distinct patrons checking out a title during a required date range, with an optional library filter. Checkout count is also returned as secondary context.
Replace the original acceptance criteria	Adopted	Section 10 uses observable setup/action/outcome tests, including valid and unavailable checkout, cross-library return, duplicate return, unauthorized access, report fixtures, reconciliation, and audit immutability.
Add auditability and recovery behavior	Adapted	Successful state changes create append-only audit evidence, while transactions roll back partial work and reject invalid or repeated requests. Persistent failed-attempt logging is excluded because authentication/application logging is outside the database scope and rolled-back requests must not create false circulation history.
Preserve completed history	Adopted	Loans are closed rather than overwritten, inventory corrections are stored as immutable events, and audit and adjustment history cannot be silently updated or deleted.
Remove the supplier assumption	Adopted	The unrelated supplier statement was removed. Purchasing and supplier management are explicitly out of scope.
Use consistent patron terminology	Adopted	`Reader` and `Customer` were replaced by `Patron` throughout the final requirements and data model.
Resolve contact-data inconsistency	Adopted	The FRD states that every identity and contact value used for implementation and demonstration is synthetic. Real personal data and credentials are excluded.
Define library status	Adopted	Library status means operational `ACTIVE` or `INACTIVE`. Inactive libraries retain history but cannot receive new operational activity.
Add a geographic map	Not adopted	"Map" was clarified as a database association between libraries and copies. Geographic interfaces, routing, and GPS are unrelated to the database deliverables.
Demonstrate security with cryptographic/dataflow analysis	Not adopted	Security is tested through roles, assignments, controlled procedures, denied access, and immutable audit records. Cryptographic/dataflow analysis is unrelated to these database workflows and outside scope.
Demonstrate ACID behavior through catalog retrieval	Not adopted	Read-only catalog retrieval does not demonstrate atomicity. Transactions are instead tested on checkout, return, donation acceptance, and reconciliation, where related rows must change together or roll back.